

Lab 17: History of Life (Module 16)

BIOL-1

Name: _____ Date: _____

Overview

Welcome to the **Deep Time Design Challenge**. You will **rank** events on an empty planet, **role-play** recovery after a mass die-off, and **argue from a tree** like a systematist. The goal is **sequential thinking** and **testable claims** about ancestry—not memorizing every million-year label.

Learning Objectives

By the end of this lab, you should be able to:

1. Order major milestones from **Earth formation** through **land colonization** with justified reasoning.
2. Connect **mass extinction** to **adaptive radiation** using niche space.
3. Read a simple **phylogenetic tree** for closest relatives and monophyly.
4. Distinguish **shared derived** vs **shared ancestral** characters.

Materials Needed

- Pencil or pen
- This worksheet

Part 1: Empty Planet — Build the Timeline Yourself

Do not look up dates. Use logic only first.

Event cards (shuffle mentally—then number 1 = earliest through 7 = latest):

- Formation of planet Earth
- First prokaryotic cells

- Oxygen accumulates in atmosphere (oxygen revolution)
- First eukaryotic cells
- Multicellular life originates
- Cambrian explosion (many animal body plans)
- Lineages colonize land (fungi, plants, animals)

1. Your predicted order (write 1–7 next to each event before you compare to Part 2 check):

Event	Your order (1–7)
Formation of Earth	
First prokaryotes	
Oxygen revolution	
First eukaryotes	
Multicellularity	
Cambrian explosion	
Land colonization	

2. Justify one hard choice: Why must **eukaryotes** come **after** oxygen trends begin to reshape metabolism and habitats (even if you are fuzzy on exact years)?

3. Instructor check (after your attempt): If your order disagrees with lecture, **what single relationship** did you fix?

Part 2: Planet Nine — Extinction Then Radiation

Science fiction scenario (biology is real):

Year 0: A global crisis wipes out **90%** of species in many habitats. Forest canopies collapse; burrowing animals survive unevenly; photosynthetic biomass crashes then slowly rebuilds.

4. Prediction: Immediately after the crisis, is **competition** for resources higher or lower for the surviving lineages in many empty habitats? **Higher / Lower** — explain.

5. Adaptive radiation: Describe **two different empty niches** survivors might diversify into (be specific—feeding mode, habitat layer, time of activity).

1.

2.

6. Earth parallel: Name **one** real mass extinction nickname + **one** group that radiated afterward (sentence each).

7. Hypothesis test: Why would **molecular clocks** and **fossils** sometimes disagree on dates —without throwing out evolution? (2–3 sentences)

Part 3: Tree Puzzle — Who Is Closest to Whom?

Consider this hypothesis tree (read tips as species; nodes are common ancestors):

Salmon and **Lungfish** share the more recent ancestor **Node X**. **Shark** joins the tree at the older **Node Y** (the common ancestor of all three). In other words: Salmon and Lungfish are **sister lineages** relative to Shark.

8. Closest relatives: Who is Salmon's closer relative—**Lungfish** or **Shark**? **Why** (use **most recent common ancestor** language)?

9. Monophyly: Is the group **{Salmon + Lungfish}** a **clade** (monophyletic) on this diagram? **Yes / No** — one sentence.

10. Characters:

Statement	Shared derived or shared ancestral?
Bony internal skeleton (shared by Salmon & Lungfish but not cartilaginous Shark in this toy diagram—treat as given)	
Vertebral column (present in all three groups)	

11. DNA updates trees: Give **two** reasons scientists revise branching order when new genomes arrive.

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Part 4: Exit Ticket — Hypotheses, Not Dogma

12. In science, why is every published tree still a hypothesis? (3 sentences)

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13. Synthesis story (5–6 sentences): Tell a **coherent narrative** from **one** ancient extinction through **one** radiation, then end with how **phylogenetics** keeps the story honest.

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